

2 *Daphnia magna* is a small invertebrate that is found in many fresh water ecosystems.

Figure 2.1 shows one *D. magna*, as seen with a light microscope. The body of *D. magna* is transparent, and the beating of the heart can be easily observed.

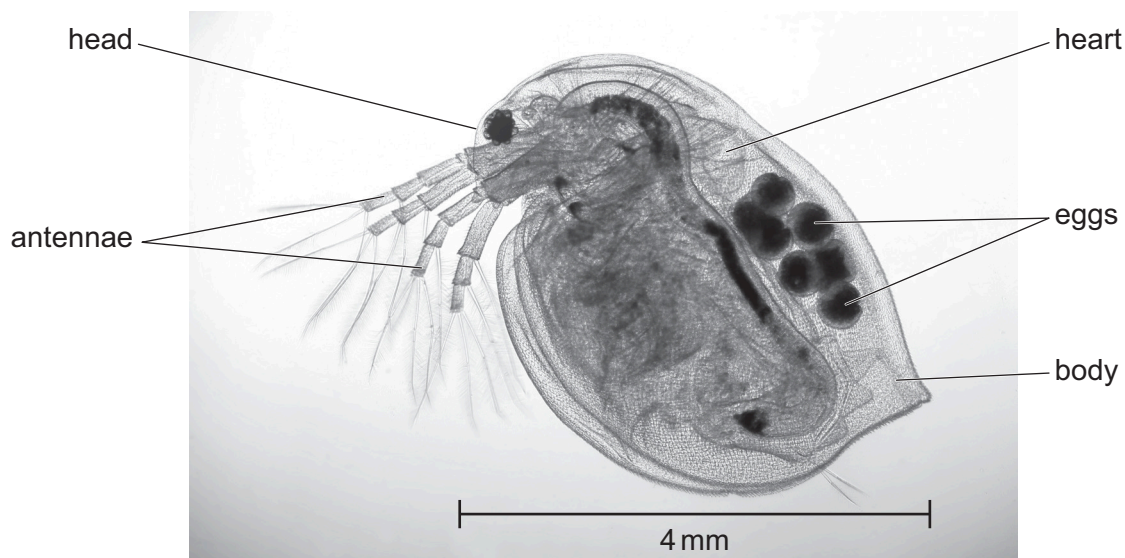


Figure 2.1

A student obtained a sample of pond water containing several *D. magna*.

To measure the time taken for one cardiac cycle in *D. magna*, the student:

- used a pipette to place one *D. magna* in the centre of a Petri dish
- added a small amount of cotton wool to reduce the movement of the *D. magna*
- added two drops of pond water to the *D. magna*
- observed the heart of the *D. magna* using a light microscope
- counted the number of cardiac cycles (heart beats) in one minute.

(a) In a trial experiment at 20 °C, the student counted 232 cardiac cycles in one minute.

Calculate the time taken for one cardiac cycle.

Give your answer in seconds and to **two** significant figures.

time for one cardiac cycle = s [1]

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

(b) Caffeine is a drug found in tea and coffee.

The student decided to investigate the effect of caffeine concentration on the time taken for one cardiac cycle in *D. magna*.

- (i) The student was given a 0.5% stock solution of caffeine from which to prepare a range of caffeine solutions of known concentrations to use in this investigation.

The student made **20 cm³** of each solution.

Describe a method the student could use to make a **proportional** dilution of the 0.5% stock solution of caffeine to obtain a 0.3% solution of caffeine.

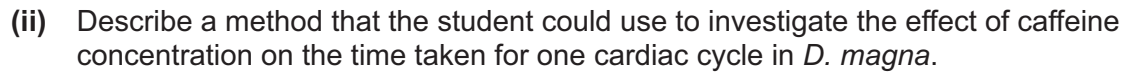
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You can assume that a detailed risk assessment has been prepared.

[7]

[Total: 10]